Tittle:-AI-Based StudyApp

An AI-powered study companion designed to transform how students learn, revise, and retain knowledge — making every study session smarter, faster, and more effective.

AI-POWERED LEARNING

STUDENT PRODUCTIVITY



The Problem We're Solving

Information Overload

Students are overwhelmed by vast amounts of content with no intelligent way to prioritise or digest it efficiently.

Passive Learning

Traditional study methods encourage rote memorisation rather than deep understanding and long-term retention.

Lack of Personalisation

One-size-fits-all resources fail to adapt to individual learning styles, pace, and knowledge gaps.



Target Users & Use Cases



Who We Serve

University Students

Preparing for exams, managing heavy reading loads, and seeking conceptual clarity.

High School Learners

Building foundational knowledge and developing effective study habits early.

Lifelong Learners

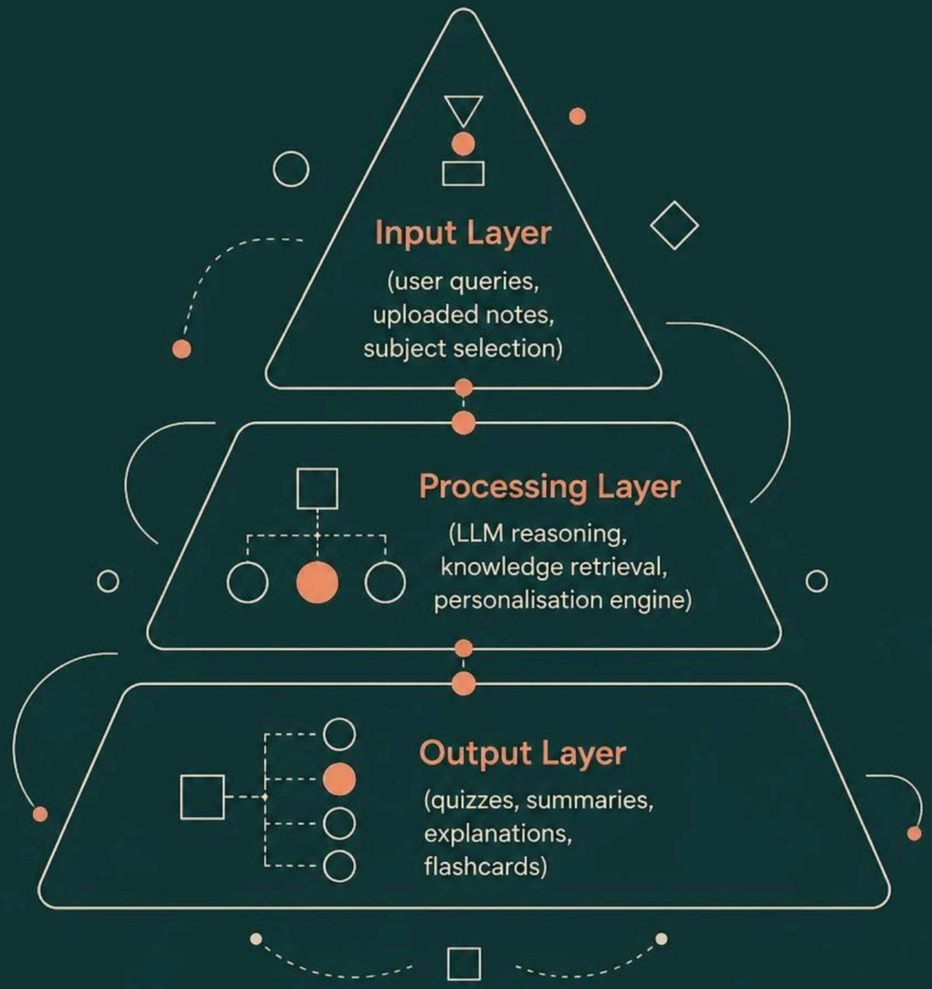
Professionals upskilling or exploring new subjects at their own pace.

The AI Engine

LLM MODEL

Why a Large Language Model?

StudyApp leverages a state-of-the-art LLM to power intelligent interactions — from generating personalised quizzes to explaining complex concepts in plain language.



- **Natural language understanding** for conversational Q&A
- **Context-aware responses** tailored to the user's subject
- **Adaptive difficulty** based on performance signals

API Integration

LLM API

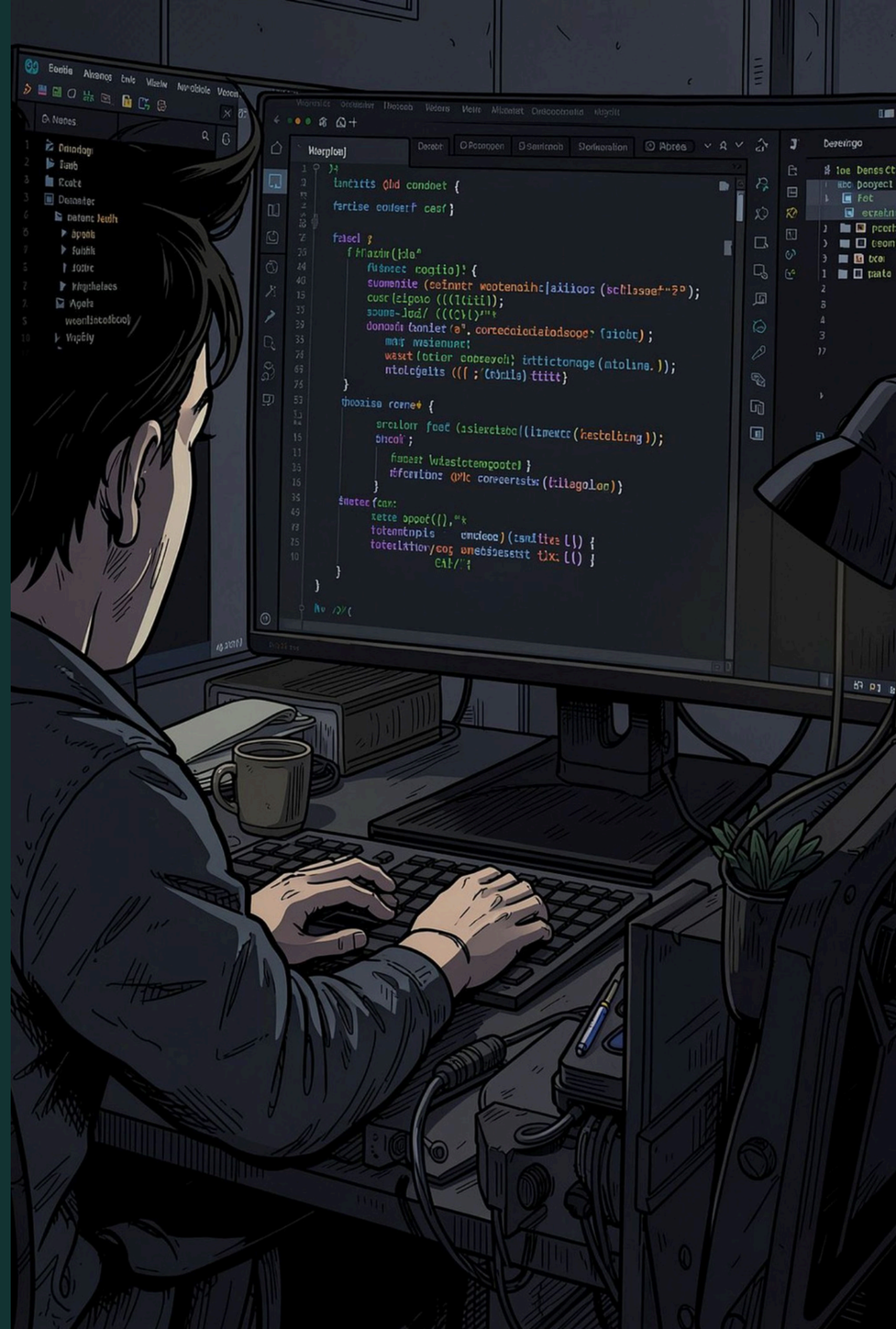
Secure REST API calls to Groq's LLM provider for real-time text generation of structured notes, flashcards, and quiz questions — with responses streamed back to the frontend as they're generated..

Secure Backend

A FastAPI server handles all API requests, keeping the provider's API key server-side at all times via environment variables — the key never reaches the browser or the frontend code.

Containerized Deployment

The application is packaged with Docker and deployed on AWS App Runner, giving it a scalable, publicly accessible HTTPS endpoint without manual server management.

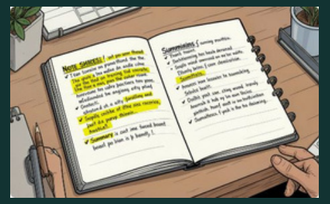


Key Features



Smart Quizzing

AI-generated quizzes adapted to your study material and performance history.



Instant Summaries

Upload any document and receive concise, structured summaries in seconds.



Spaced Repetition

Built-in flashcards using spaced repetition algorithms for long-term retention.

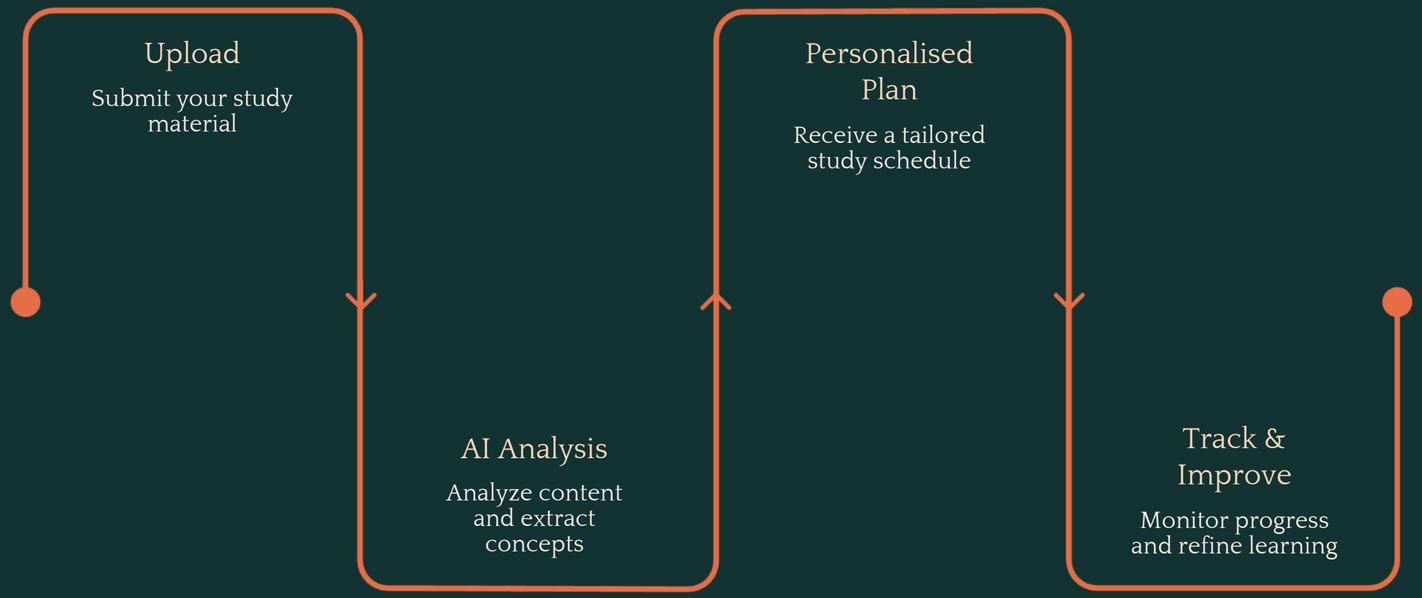


Study Chatbot

Ask anything about your subject and get clear, contextual explanations instantly.

Expected User Experience & Outcomes

The StudyApp Journey



A seamless, guided experience that turns passive reading into active learning.

Measurable Outcomes

1

Faster Revision

Students revise key concepts in a fraction of the usual time.

2

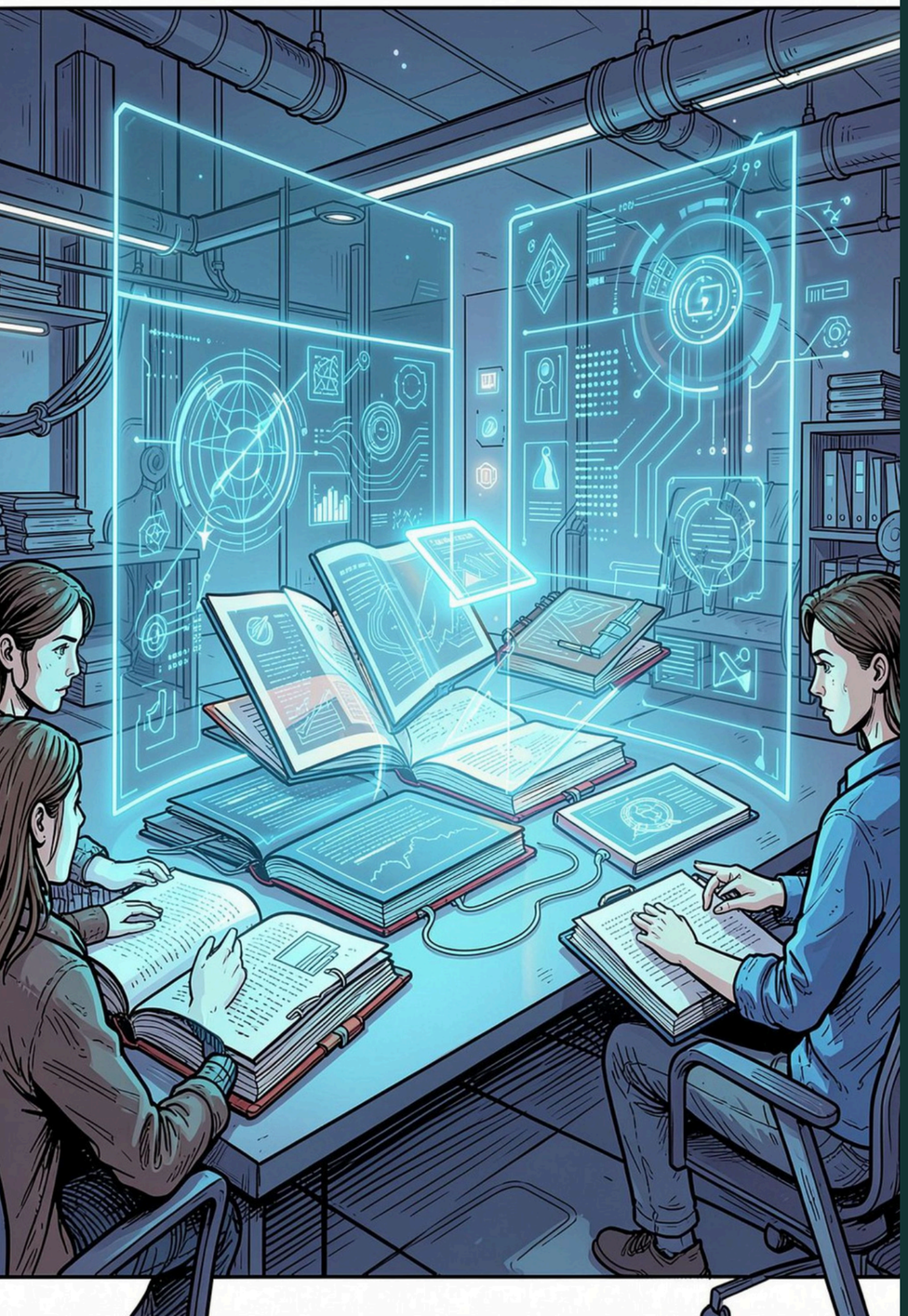
Better Retention

Spaced repetition and active recall improve long-term memory.

3

Higher Confidence

Personalised feedback reduces exam anxiety and builds readiness.



Future Scope

1

Phase 1 — MVP Launch

Core quizzing, summaries, and chatbot features for web users.

2

Phase 2 — Mobile App

Native iOS and Android apps with offline study mode.

3

Phase 3 — Collaboration

Group study rooms, shared decks, and peer-to-peer learning tools.

4

Phase 4 — Institutional

University partnerships, LMS integrations, and educator dashboards.

The long-term vision is to become the default AI study companion for learners worldwide — accessible, adaptive, and always improving.